

PROJECT OVERVIEW

- **Project Title : AI Learning & Study Assistant**

- **Problem Statement :**

Students often need help in understanding difficult topics, planning their study time, and practicing what they have learned. Usually, these activities are done separately using different resources. This project aims to bring these basic learning needs together in one AI-based assistant.

- **Description of the Project :**

The AI Learning & Study Assistant is an Agentic AI application developed to help students during their learning process. It can answer questions from a study textbook, create study plans, and generate practice quizzes.

The project uses **RAG, LangGraph, ChromaDB** and **Groq LLM** to make the assistant more useful and interactive. Instead of giving the same type of response for every question, the AI agent decides what action is needed and uses the suitable tool.

OBJECTIVES & PROPOSED SOLUTION

- **Project Objectives :**

The main objectives of this project are:

1. To build an AI assistant for student learning.
2. To answer questions using information from a textbook.
3. To generate simple and personalized study plans.
4. To help students practice through quizzes.
5. To use an agent-based workflow for selecting the right tool.
6. To maintain the conversation context during interaction.

- **How the Agentic AI Solution Works :**

The system works in a simple flow. First, the student enters a question or request. The LangGraph agent reads the request and decides what type of help is required.

For textbook questions, it uses the **RAG search tool** to find relevant information from the stored textbook content. For study-related requests, it can generate a study plan, and for practice requests, it can generate a quiz.

Finally, the Groq-based LLM uses the available information and gives the response to the student.

- **Basic workflow:**

Student → AI Agent → Select Tool → Process Request → Generate Response

- **Key Features :**

Textbook Question Answering:

The assistant searches the textbook content and provides answers based on the retrieved information.

Study Plan Generation:

Students can ask for a study plan for a particular topic and number of days.

Practice Quiz:

The assistant can generate quiz content for practice and self-assessment.

Agentic Behaviour:

The system can choose the required tool based on the student's request.

Conversation Memory:

The project uses LangGraph memory to maintain the conversation state.

IMPLEMENTATION & RESULTS

- **Technologies / Tools Used :**

1. **Python** – Main programming language
2. **LangGraph** – Agent workflow and memory
3. **LangChain** – AI and tool integration
4. **Groq LLM** – Response generation
5. **ChromaDB** – Storing and retrieving document embeddings
6. **Hugging Face Embeddings** – Converting text into vectors
7. **PyPDF** – Reading textbook PDF files
8. **Rich** – Terminal interface and output formatting

- **Working Process :**

The project first loads the textbook PDF and extracts its content. The text is divided into smaller sections and converted into embeddings. These embeddings are stored in ChromaDB.

When the student asks a question, the system searches the stored content and retrieves the most relevant information. The AI agent then uses the appropriate tool and the LLM generates the final response.

For example, when a student asks for a **5-day RAG study plan**, the agent identifies that a study plan is required and generates a day-by-day schedule.

- Screenshots / Output :

Figure 1. AI Learning & Study Assistant Initialization

The first screenshot shows the application starting successfully. The system loads the required model, indexes the textbook content into ChromaDB, and displays the available features.

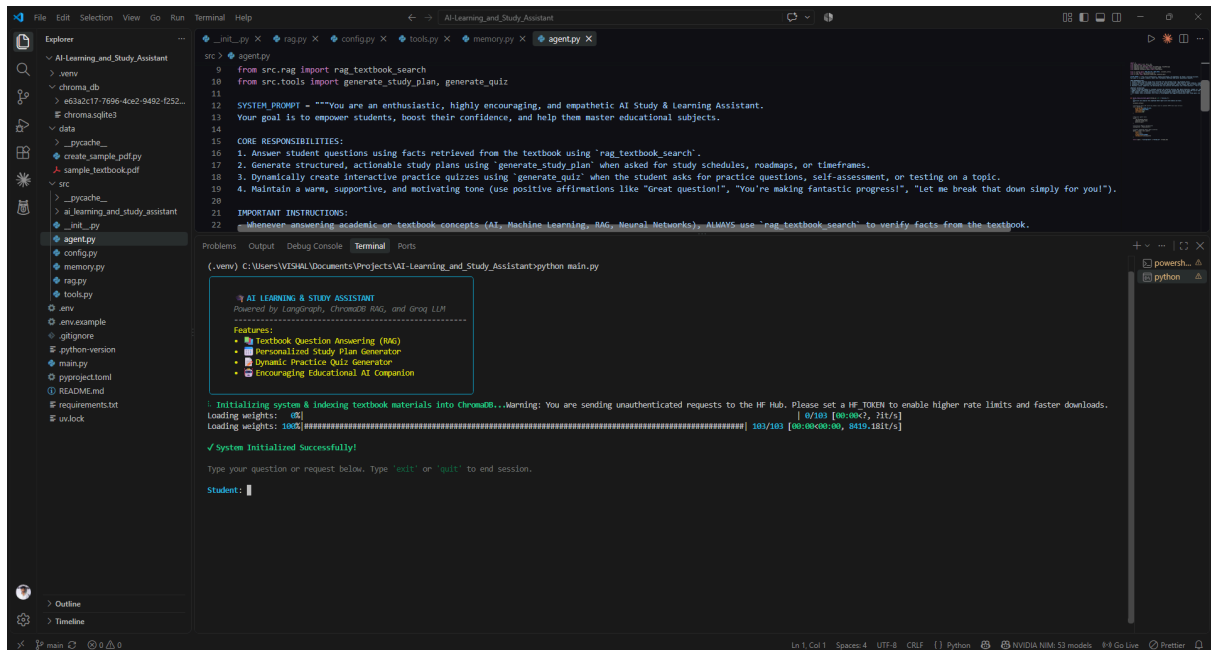
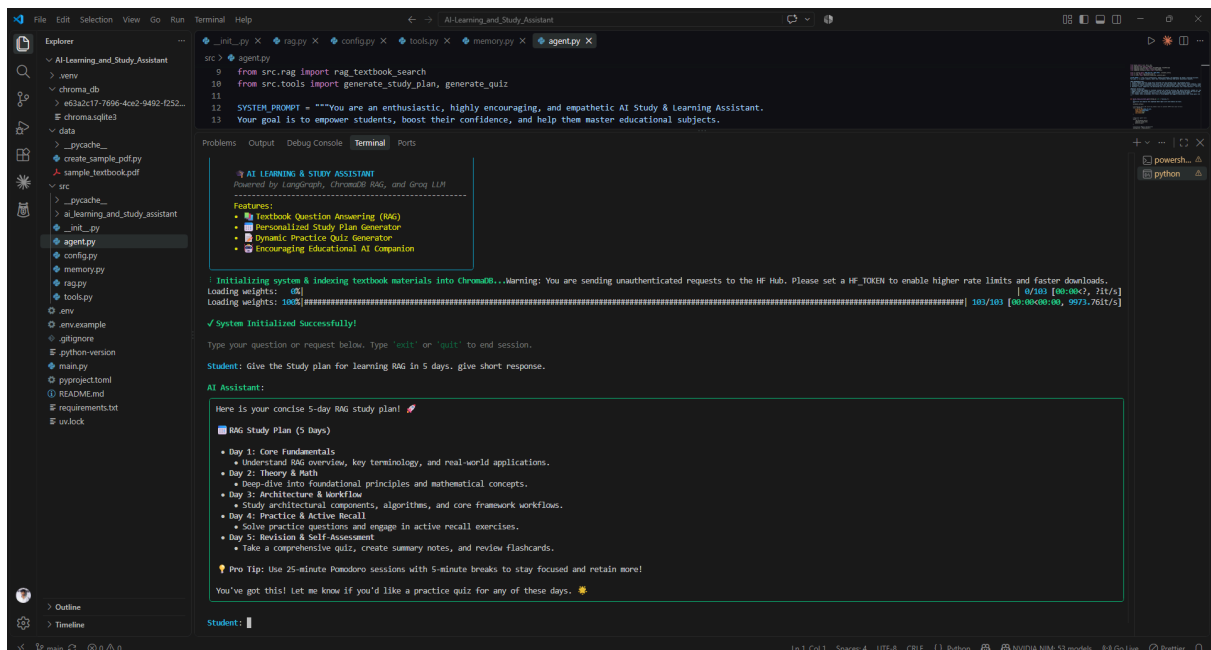


Figure 2. Study Plan Generation

The second screenshot shows the assistant generating a **5-day study plan for learning RAG**. The plan contains daily topics and also gives a Pomodoro-based study suggestion.



- **Results Achieved :**

The project successfully demonstrates the main idea of an Agentic AI-based learning assistant. It can retrieve textbook information, generate study plans, and support quiz-based learning through an interactive interface.

The working output also shows that the different components such as **RAG, tool calling, memory and LLM** can be combined into one useful student-support system.

CONCLUSION & FUTURE SCOPE

- **Project Conclusion :**

The **AI Learning & Study Assistant** is a simple but useful application of Agentic AI in education. The project combines different AI technologies to help students with studying, understanding concepts and practicing questions.

One of the main advantages of the project is that the agent does not follow a single fixed response pattern. It can decide whether the student needs textbook information, a study plan or a quiz and then use the appropriate tool.

- **Challenges Faced :**

During development, integrating different AI components such as LangGraph, RAG, ChromaDB and the LLM was one of the main challenges. Preparing the textbook data for retrieval and managing the agent workflow also required testing and debugging.

- **Future Enhancements :**

In the future, the project can be improved by adding a **web interface, student progress tracking, better quiz evaluation, multiple textbook support, voice interaction and personalized learning recommendations.**

- **References :**

1. LangGraph Documentation
<https://langchain-ai.github.io/langgraph/>
2. LangChain Documentation
<https://python.langchain.com/>
3. ChromaDB Documentation
<https://docs.trychroma.com/>